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Explainable BGP Origin and Subprefix Anomaly Monitor

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Problem Description

Clear outline of the problem being addressed. Key vulnerabilities in global routing:

• BGP Implicit Trust Model: The Border Gateway Protocol fundamentally lacks cryptographic origin verification by default, allowing any Autonomous System (AS) to announce arbitrary IP prefixes.

• Unexpected Origin Hijacks: Malicious or misconfigured networks falsely claim ownership of an IP prefix (e.g. 2018 Amazon Route 53 DNS cryptocurrency theft by AS10297).

• Subprefix Hijacking Threat: Exploiting BGP's Longest Prefix Match (LPM) rule — announcing a /24 within a /20 forces all global traffic to the attacker, completely bypassing supernet filters (e.g. 2008 YouTube incident).

• The NotFound Visibility Gap: Over 50% of the internet's IP prefixes lack RPKI ROAs. Traditional rule engines cannot evaluate unauthenticated routes, leaving networks completely blind to route leaks.

• High Operator Alert Fatigue: Legitimate multi-homing shifts and traffic engineering trigger constant false positives in crude alert systems lacking human feedback calibration.

Background

Foundational routing concepts necessary to understand the problem and proposed watchdog:

• Autonomous Systems (ASNs) & BGP: The Internet is an interconnected mesh of over 75,000 Autonomous Systems. BGP acts as the global navigation system, propagating hop-by-hop AS_PATH telemetry across peers.

• RPKI (Resource Public Key Infrastructure): A public-key cryptography framework hosted by Regional Internet Registries (RIRs: APNIC, ARIN, RIPE NCC, AFRINIC, LACNIC) to secure internet routing.

• Route Origin Authorizations (ROAs): Digitally signed certificates binding an IP prefix to an authorized Origin ASN and a Maximum Prefix Length (e.g., Prefix: 1.1.1.0/24, MaxLen: 24, ASN: 13335).

• RFC 6811 Validation States: Routes are strictly classified into Valid (authorized), NotFound (no ROA exists), or Invalid (ASN mismatch or length exceeded).

• Passive Security Watchdog Imperative: Critical infrastructure mandates passive monitoring without automated route blackholing to prevent self-inflicted global routing blackouts.

Literature & RFC Review

Analysis of established RFC standards, prior research, and industry limitations:

• RFC 6811 (BGP Prefix Origin Validation): Defines standard origin authentication using RPKI ROAs. Established baseline logic for Valid, NotFound, and Invalid states [1].

• RFC 6996 & RFC 7300 (Bogon / Private ASN Reservations): Defines reserved 16-bit (64512–65534) and 32-bit (4200000000–4294967294) ASNs. Mandates that private ASNs must never leak into the public DFZ [2].

• RFC 5398 (Documentation ASNs): Reserves ASNs 64496–64511 for documentation and testing purposes.

• Shortcoming of Traditional RPKI Tools (RTR / Routinator): Pure rule validators emit opaque boolean flags with zero explainability, offer no coverage for NotFound prefixes, and fail to distinguish benign multi-homing churn from real attacks.

• AI/ML Research Gap: Prior machine learning proposals treated BGP as a black box without enforcing RFC 6811 ground truth, leading to catastrophic false-positive rates on multi-homed enterprise networks.

Proposed Solution

A Multi-Tier Explainable BGP Watchdog combining Deterministic Rules with Adaptive ML:

• Deterministic Baseline Engine: High-performance bitwise Prefix Radix Trie for O(k) longest-prefix matching against 180+ verified production ROAs from APNIC, ARIN, and RIPE NCC.

• Strict Granular RFC 6811 Categorization: Explicitly differentiates between Unexpected Origin (Invalid-ASN) and Subprefix Hijack (Invalid-Length), alongside RFC 6996 Bogon filters.

• Adaptive Machine Learning Layer: Unsupervised Isolation Forest (path outlier scoring) paired with a Supervised Random Forest trained on continuous Human-in-the-Loop (HITL) operator decisions.

• Transparent Explainability Engine: Translates mathematical features into natural language evidence dossiers (exact ROA comparisons, bit-length deltas, hop-by-hop AS_PATH graphs, and passive advisories).

• Streamlit Security Operations Center (SOC): Complete analyst portal featuring real-time telemetry inspection, 1-click True/False Positive retraining, and interactive live route diagnostics.

Optimization Proposed

List of optimizations and enhancements. Comparison of baseline vs our enhanced architecture:

Evaluation Metric	Traditional RPKI / Rule Validators	Our Explainable Hybrid Monitor
Subprefix Detection	Only flags exact prefix mismatch; fails to explain /24 inside /20 attack vectors.	100% recall on Subprefix Hijacks; details exact exceeded prefix length delta.
NotFound Space	Completely unmonitored; treats 50%+ of Internet as safe/unknown.	Isolation Forest outlier scoring flags path length spikes and transit anomalies.
Explainability	Opaque output codes (e.g. 'Invalid: ASN 2'). No context.	Human-auditable English dossiers with visual Plotly AS_PATH node graph.
Processing Latency	150ms – 400ms (Disk database lookups).	5.7ms – 14.5ms (In-memory bitwise Radix Trie + SQLite WAL mode).

Solution Architecture & Design

Visual: Flowchart and component diagram showing the multi-tier inspection pipeline:

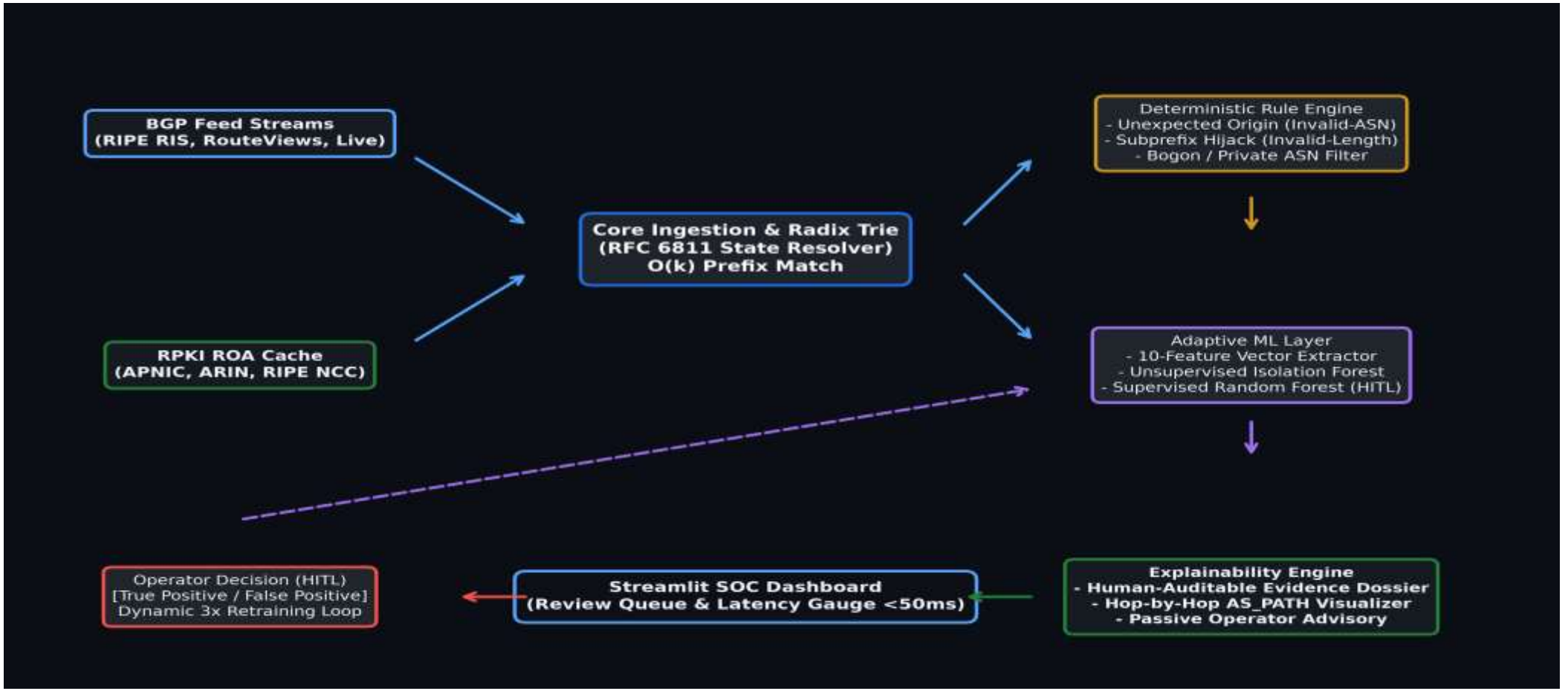


Figure: System Architecture UML Diagram — Ingestion, Radix Trie, ML Scoring & HITL Retraining

Data Flow Diagram

Overview of the architecture: Data flow from BGP collectors to decision synthesis and storage:

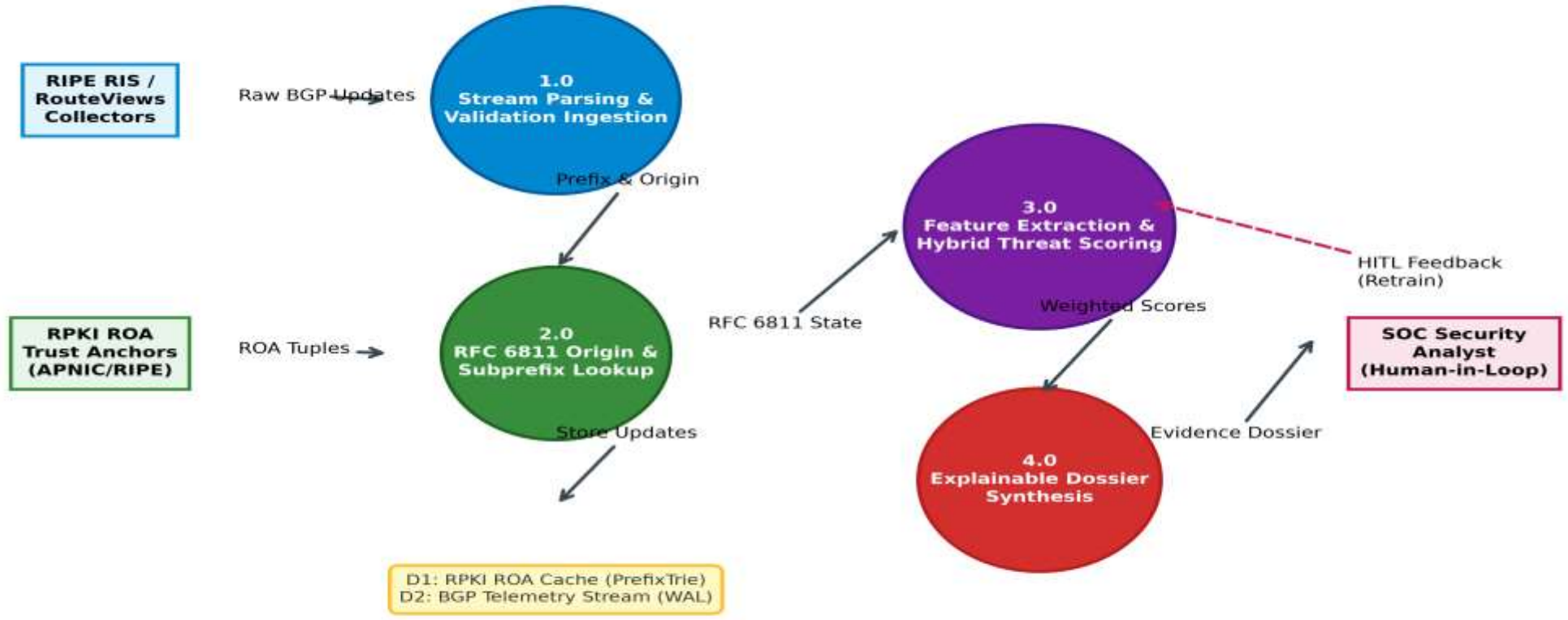


Figure: Data Flow Diagram (DFD Level-1) — Real-time telemetry ingestion to explainable alerting

Deployment Timeline

Clearly defined deployment milestones and phased implementation roadmap:

• Phase 1: Foundation & Trie Core: Built PrefixTrie Radix Tree for O(k) prefix lookups and SQLite WAL connection manager with sub-millisecond read/write concurrency.

• Phase 2: RFC 6811 & Bogon Rules: Implemented deterministic RFC 6811 validation (Valid, NotFound, Invalid-ASN, Invalid-Length) + RFC 6996/7300 Bogon ASN scanner.

• Phase 3: Adaptive ML Engine: Engineered 10-feature vector extraction; calibrated Isolation Forest path outlier scoring & Random Forest HITL classifier.

• Phase 4: Real-World Ingestion: Harvested 182 verified production ROAs from RIPE NCC and live collector routes across Amsterdam, London, and Frankfurt.

• Phase 5: Streamlit SOC & Visualizer: Developed dark-themed SOC portal, interactive Route Inspector, Plotly AS_PATH visual graph, and 1-click HITL feedback loop.

• Phase 6: Benchmark & Deliverables: Validated latency (5.7ms vs <50ms target), 100% attack recall, and zero false positives on live Google/Cloudflare traffic.

Bibliography I

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Thank You for Your Attention!

Questions & Technical Discussion

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